

# ***Online Retail Sales Analysis***

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***Tools Used :*** Excel,SQL,Power Bi

***Dataset :*** UCI Online Retail Dataset

## **Objective :**

The objective of this project is to analyze online retail sales data and identify sales trends, profitable products, customer behavior and business opportunities through an interactive Power Bi dashboard.

## **Business Problem :**

*The company wants to understand :*

- I. Which product generate the highest revenue?*
- II. Which months perform best?*
- III. Which countries contribute the most sales?*
- IV. What are customer purchasing patterns?*
- V. How can profitability be improved?*

## ***Dataset Information :***

### ***Dataset Source :***

UCI Machine Learning Repository –Online Retail Dataset.

### ***Dataset Description :***

The dataset contain transactional records of a UK based online retailer between December 2010 and December 2011.

### ***Dataset Fields :***

| Column      | Description        |
|-------------|--------------------|
| InvoiceNo   | Invoice Number     |
| StockCode   | Product Code       |
| Description | Product Name       |
| Quantity    | Quantity Sold      |
| InvoiceData | Transaction Date   |
| InvoiceTime | Transaction Time   |
| UnitPrice   | Product Price      |
| CustomerId  | Customer Id        |
| Country     | Customer's Country |

### ***Total Records :***

Original Records=536642

After cleaning =522908

## ***Data Cleaning Process :***

### ***Data Cleaning Steps :***

Data cleaning process is done in excel.

- Removed duplicate records
- Removed cancelled orders
- Handled missing customer id's
- Corrected data format
- Create different time column
- Handled missing stockcode and invoice no

### ***Revenue Formula :***

Revenue = Quantity\*Unit Price

Some of the data related process is also done on Power Bi 's power query.

## ***Tools and Technologies :***

### ***Excel***

#### **Used for :**

- Initial data inspection
- Data cleaning

### ***SQL***

#### **Used for :**

- Data validation
- Aggregations
- Business queries

### ***Power Bi***

#### **Used for:**

- Dashboard creation
- KPI cards
- Interactive visualizations
- Dax measures

## ***KPI Metrics :***

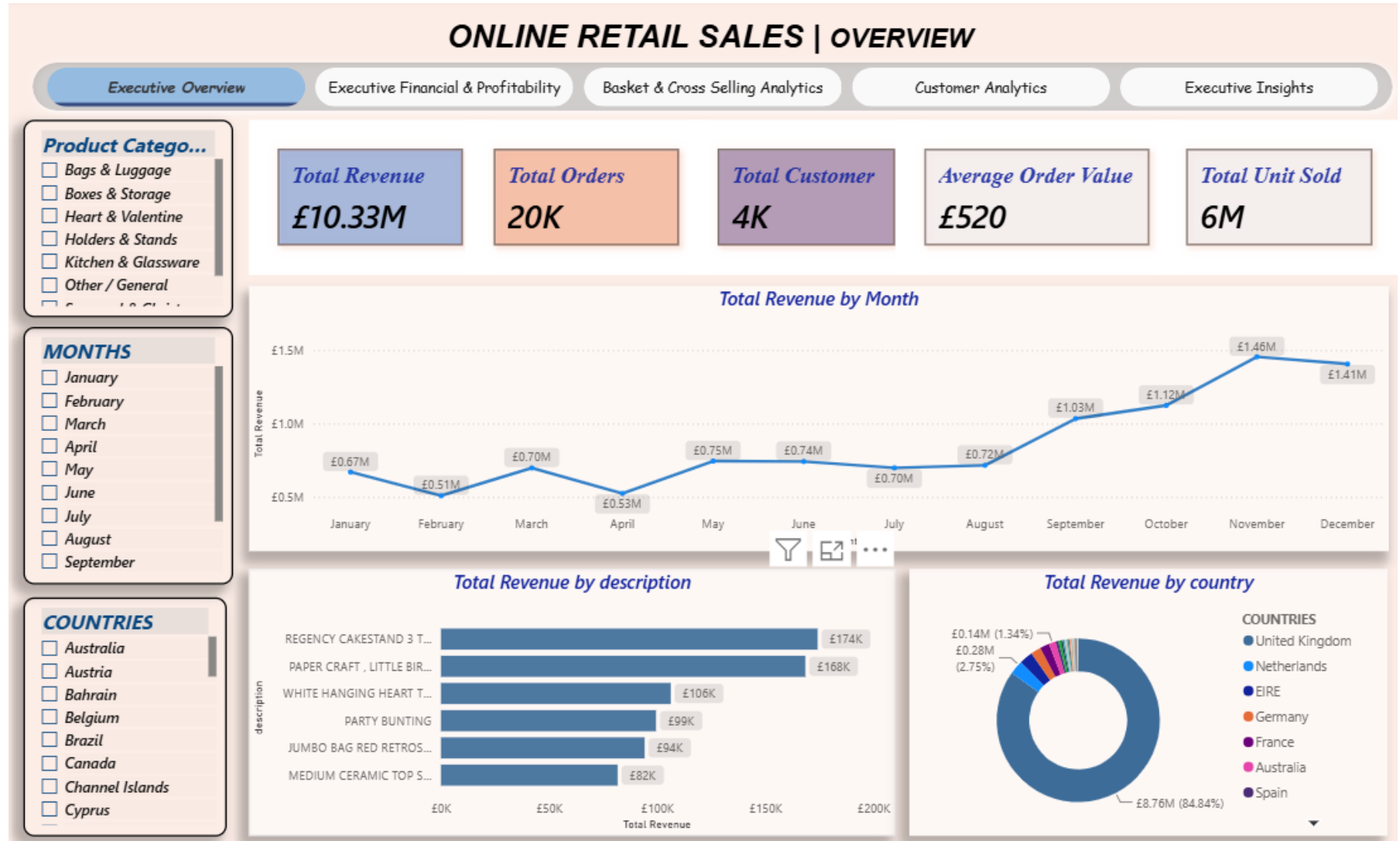
| <b>KPI</b>          | <b>Descriptions</b> |
|---------------------|---------------------|
| Total Revenue       | Total sales amounts |
| Total Orders        | Total Transactions  |
| Total Customers     | Unique customers    |
| Average Order Value | Revenue / Orders    |
| Gross Profits       | Total Profits       |
| Profit Margin       | Profit / Revenue    |

# Dashboards Pages :

## 1.Executive Overview

### Contains:

- Revenue KPI
- Orders KPI
- Customers KPI
- Revenue trend
- Country analysis
- Product analysis



# Executive Financial & Profitability :

## Contains:

- Revenue
- Profits
- Profits Margin
- Average Order Value
- Monthly Profits Trends

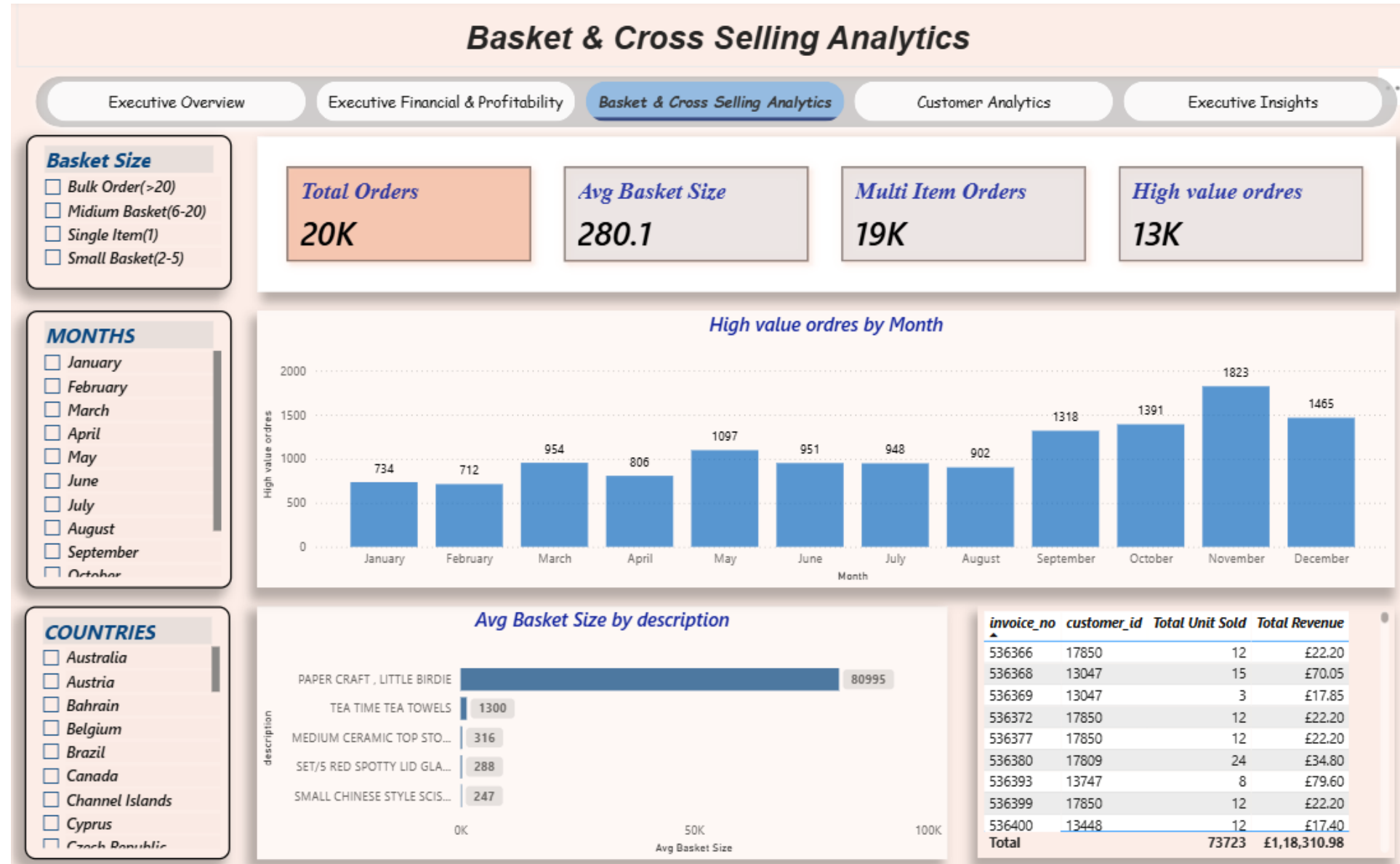




# Basket & Operational Insights

## Contains:

- Average Basket Size
- Multi-item Orders
- High-value Orders
- Basket Trends



# Customer Analytics :

## Contains :

Total Customers

Revenue Per Customer

Customer By Months



# ***Key Insights***

## ***Sales Insights***

- Total Revenue reached 10.33M.
- November generated the highest monthly revenue.
- United Kingdom contributed most sales.
- Top-selling products generated significant revenue concentration.
- Q4 showed the strongest sales performance.

## ***Customer Insights***

- Multi-item orders represented a large portion of transaction.
- Average order value remained stable throughout the year.

## ***Product Insights***

- A small number of products generated a large percentage of revenue.
- Seasonal product performed strongly during peak months.

## ***Recommendations :***

1. Increase inventory before Q4.
2. Focus marketing on high-performing product.
3. Improve customer retention programs.
4. Expand sales efforts outside the UK.
5. Promote bundled purchases to increase basket size.

## ***Conclusion :***

This project successfully analysed online retail sales data using Excel,SQL,and Power Bi. The dashboard provides valuable insights into revenue trends ,customer behaviour , product performance and profitability ,helping businesses make data-driven decisions.